Extract Blade Violates Secure Shot: Exquisite Corpse as a Speculative Design Method

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Abstract

What if sound waves could protect your privacy and vaccines could protect your biometric data? What if you could watermark your voice and trees could store data? These are not questions, but ideas emerging from playful speculative design workshops. By drawing on the surrealist exquisite corpse technique, randomly combined sentences prompted exploratory discussions in which concepts from various technological fields were imaginatively linked to cybersecurity and privacy for visually impaired and blind elderly. The results showed that, exquisite corpse holds methodological potential, as the unpredictable nature of the technique can stimulate unconventional thinking, which can be applied in various domains.

CCS Concepts

• **Security and privacy → Social aspects of security and privacy**.

Keywords

Exquisite Corpse, Speculative Design, Privacy, Security, Assistive Technology, Elderly, Visual Impairment

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1 Introduction

As digital technologies become increasingly embedded in everyday life, they also introduce complex and often unintended security and privacy implications. Emerging technologies are similarly difficult to reason and conjecture about by using conventional evaluation methods, as their risks may not become apparent until they are situated in real life contexts. These risks include, but are not limited to, security and privacy risks such as surveillance, vandalism, data theft, and more [2, 6, 8, 12, 16, 21, 22]. These challenges motivate methodological approaches that enable researchers, designers, and policy makers to think beyond current realities and anticipate how future technologies might shape, and be shaped by human interactions. Speculative design practices have previously been used to support such prediction attempts, including approaches such as design fiction to anticipate future challenges of software development [15], the data economy in 2037 [19], and functional goals associated with well-being technology [1]. Yet, these speculations are often implicitly constrained by the limits of the human imagination, including plausible and probable futures. To push beyond these boundaries, a method must allow itself the freedom to break from the constraints of human imagination by embracing odd, exaggerated, or seemingly illogical outcomes that emerge by chance rather than intentional speculations.

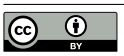
In this extended abstract, we present a short experimental study that introduces a speculative design method by using the surrealist *exquisite corpse* technique. The exquisite corpse can be seen as a sketching technique, where randomly combined input can be a catalyst and a creative way to spark ideas, strange combinations, and provide new views and interpretations of possible real scenarios [9]. We demonstrate the applicability of the method through an experimental study focused on assistive technology (hereafter AT) for blind or visually impaired older adults as a use case, as a growing aging population and age-related health complications, such as vision loss, place increasing pressure on the welfare systems, and ATs are often proposed as one way to alleviate this burden [14, 17, 18, 24]. The contributions of this study are twofold; First, we contribute to the conversation about the future of AT by speculating on the security and privacy implications that might be overlooked or under-assessed in conventional design and development processes, and second, we examine the methodological potential of exquisite corpse as a research and speculative design tool to be applied to other kinds of technology in general.

Our findings suggest that the exquisite corpse holds considerable potential for generating deliberately far-fetched ideas, such as using vaccines to protect biometric data or treating scale itself as a security mechanism. Rather than serving as final solutions, these ideas act as points of departure and open speculative pathways which can be followed towards more realistic and actionable solutions.

2 Background & Related Work

In this section we outline the background and briefly highlight some prior research that have employed similar methodological approaches.

Speculative design – a branch of critical design – imagines possible future scenarios to examine social, political, technological, and ethical challenges, and to inspire new ideas and solutions. Speculative design does not aim to predict the future, but to expand the realm of possibilities, conceptualized through probable futures, possible futures, plausible futures, and preferable futures [11, 20]. Within the domain of technology, speculative design has been used



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to reverse the usual order of development, so instead of new technology being developed and then assessing their impact, design speculations can help drive responsible innovation as well as open up new opportunities that are liberated from current constraints. Similarly, within the domain of privacy speculative design efforts has been utilized for political and social provocation to raise awareness about issues emerging from technological advancements, such as surveillance, manipulation, deep-fakes, etc. [20]. Loureiro-Koechlin et al. used a related design practice, design fiction, to elicit security risks in software development through a collaborative setup involving security experts, science fiction enthusiasts, and students. Among their findings was that security experts contributed the least to identifying prospective threats and that the creative writing process opened up possibilities to imagine unconventional threats – mainly those in the socio-political/safeguarding category [15]. Similarly, Riss et al, employed design fiction to anticipate the challenges of data economy in 2037, and their prospective solutions [19].

In 1925, a group of surrealist artists invented a game called Exquisite Corpse (French: "*Le cadavre exquis*"), originally played as a word-based parlor game, the game was designed to harness the creative powers of the human imagination by relying on collaboration, chance, and unexpected juxtaposition [7, 10, 23]. The goal is to keep the creators' conscious choices to a minimum [9]. The extent to which exquisite corpse has been explored as a research method remains limited. Some prior research have explored the exquisite corpse as a methodology in various disciplines, but to the best of our knowledge, none has used it within security and privacy. Dasjardins and Wakkary [10] conducted a study employing exquisite corpse in video format to explore low-level physical interactions in interaction design, concluding that this method could well be used to sketch and explore other types of interactions and surprising combinations. De la Fuenta [9] reports on exquisite corpse used to address relevant questions for global health, such as highlighting mortality associated with the COVID-19 pandemic. Similarly, in neuroscience, researchers took the exquisite corpse and turned it into a neuroscience experiment to study how the brain works during creative improvisations. This method was chosen because it is collaborative and adaptable across many art forms [7].

3 Methodology

3.1 Exquisite Corpse

The game is played by at least three participants and is accessible to anyone. Using either words or drawings, one player begins by making a contribution (e.g., writing a definite or indefinite article and a noun) then folds the paper to conceal their input before passing it to the next player, who adds their own contribution (e.g., a verb). This process continues until all participants have contributed. Historically, the standard sequence for word-based play was: "definite/indefinite article + adjective" ⟶ "noun" ⟶ "verb" ⟶ "definite/indefinite article + adjective" ⟶ "noun." Once everyone is done, the paper is unfolded to reveal the mash-up of words – players may agree to make minor changes to ensure grammatical consistency [5, 9]. The combination of contributions from this sequence gave rise to the game's original name *"The exquisite corpse shall drink the new wine."* (French: "*Le cadavre exquis boira le vin nouveau*") [5].

While the final results of this game might not make much sense as a whole, it can be seen as a way to find unexpected combinations that yield insights into our unconscious or dream worlds [23], encourage collaborations between science and art to address scientific challenges [9], as well as an example of how collective efforts can be creative [10]. In this study, exquisite corpse is a way to envision the implications of security and privacy in the deepening weave of technology into the lives of older adults.

3.2 Participant Recruitment

In total, two groups of participants were recruited and the study was conducted separately with each group. The first group consisted of individuals within the HCI domain. These participants were recruited by convenience sampling and consisted of two PhD students and one Bachelor student. The second group consisted of three cybersecurity knowledgeable participants, that is either students taking an advanced cybersecurity course or people working with cybersecurity in their job, who were given the opportunity to voluntarily participate in the study. Participants were recruited from HCI and security & privacy communities to combine creative and exploratory design perspective with technical grounded security expertise. The study was announced verbally and in writing. Prior to participation, all participants were explicitly informed about the purpose of the study and their voluntary participation was ensured through a consent form. Our institution does not have a mandatory Institution Review Board for studies, but we adhered to the common practices of addressing ethical considerations [4]. We did not provide any remuneration to the participants.

In total, two groups of three participants were formed, and the groups conducted the study independently of each other. The studies were carried out in November 2025, and **workshop 1** (HCI) lasted 40 minutes and 40 seconds **workshop 2** (security and privacy) lasted 46 minutes and 25 seconds. In the results section P1, P2, P3 belong to workshop 1, and P4, P5, P6 belong to workshop 2.

3.3 Study Design

3.3.1 Pre and Post-Study Survey. At the outset of the study, a short presentation was prepared to give the participants an introduction to the session and the topic, how the game is going to be played, and the chance to ask for clarifying questions. Then participants completed a short survey in which they were asked to type the first word or short phrase that came to mind when hearing "AT + security + privacy + visual impairment". The goal was to warm up and get into the mindset of the theme of the study, as well as provide the basis for the post-study survey. Their responses were collected using Mentimeter [1] to generate a collective word cloud. The same exercise was repeated after the workshop to capture potential changes in their associations. Due to length constraints of this paper, we do not report on or analyze the findings from the pre and post-study surveys.

3.3.2 Exquisite Corpse study design. After completing the pre-study survey, the games began. Participants sat around a table and one person started by writing a definite/indefinite article followed by an adjective. They then folded the paper and passed it on

[1] https://www.mentimeter.com/

to the next participant, who added a noun. The game continued in this manner according to the structure described in section 3.1. At the end of the first round, the paper was unfolded to reveal the full sentence, and minor adjustments were made (if necessary) to ensure grammatical consistency. The facilitator then asked the whole group a set of questions to assess the extent to which the sentences could be considered future realities. The questions were derived using a structured analysis technique, as outlined by the CIA's Tradecraft Primer [13]. Our short list of questions were based on the *"what if"* analysis technique, which is used to assume that an event has occurred with potential impact, be it positive or negative, and to explain how it might come about [13]. The following questions were asked and discussed after each round: (1) How do you interpret this? (2) How likely do you think this is to happen in 50 years? (3) What would trigger this event to happen?

The game was repeated twice in workshop 1 and three times in workshop 2. The workshops were audio recorded and subsequently transcribed. We describe our findings in Section 4. Given the length constraints of this paper, we do not apply formal analysis techniques but report our findings as responses to the reflective questions. Before the workshops, a pilot was conducted and changes to the workshop structure and provided instructions were made based on the pilot.

4 Results

4.1 Workshop 1

This workshop was strongly shaped by a solarpunk [2] theme, and concepts such as dependence, independence, and interdependence with nature, community and biology played a central role. The participants created a total of two sentences, and both sentences generated rich discussions and the exchange of ideas. Although explicitly informed about the topic of the workshop, participants focused less on security and privacy and more on technology and innovation for the blind and visually impaired, in general. We report on the sentences created in chronological order.

A SOFT BRAILLE DISPLAY RELIES ON A SEE-THROUGH TREE

When asked how they interpreted this sentence participants discussed an alternative future in which technology is more deeply woven into nature, and human sensory capacities are expanded, for instance by creating "vision" through other senses. Tactility was described as a privacy-enhancing alternative to auditory cues, which were seen as more vulnerable to eavesdropping in public spaces. Participants also reflected on the merging of technology and nature by rethinking what counts as technology: could a tree, for example, store data?

> "*But audio notes could be vibrations. So I think you can like feel a lot of things from audio too.*" [P3]

Triggering events for such a future rely on rethinking and reimagining technology, and on challenging the status quo, for instance, how we access information, which materials and resources we draw on, and which senses we assign importance to, e.g., for accessing information.

A WET SKIN CELL COMMUNICATES WITH ONE HEAVY COMMUNITY

This sentence was interpreted as something related to biology. "A wet skin cell" triggered discussions about alternative forms of communication, e.g., how moisture on the skin can signal anxiety or other emotions that cause sweating.

"A heavy community" was understood in multiple ways. The "heaviness" could mean burden but it could also suggest weight in the sense of strength, support, and closeness. "Community" was also read broadly, referring not only to groups of people, but it well describe the body's bacterial flora as a kind of community in its own right.

> "*This is bringing me to like, implantable. Like maybe not necessarily medical implants or something, but like biohacking and stuff like that. This one has like soft cyborg vibes.*" [P2]

The likelihood of such a future scenario was considered high, and participants noted that we are already partway there through current medical and scientific advancements, including the ability to modify cells. They pushed the idea further by exploring how biological cells might be modified and combined with technology to provide various forms of assistance. In this line of thinking, technology shifts from being something external or foreign to becoming something more integrated, natural and ultimately more personal to the individual. Triggering events for such future would rely on a more exploratory approach to healthcare.

4.2 Workshop 2

This workshop was shaped by an overall nanopunk [3] theme and balanced a mix of optimistic and pessimistic near-future scenarios. Consequently, information security and biotechnology emerged as central focal points. The participants created a total of three sentences, reported in chronological order. One participant observed that the sentences become progressively darker and more aggressive. In contrast to Workshop 1, participants in this workshop were slightly more challenged in interpreting the sentences and imagining alternative futures, and consequently, their interpretations remained closer to current realities. On the other hand, their discussions were much more specific to security and privacy as opposed to technology in general.

A NANO TABLE SPEAKS AN ELECTRIC SCHOOL

Participants interpreted this sentence as relating to education, particularly how learning could become more convenient if certain cognitive functionalities, such as skill acquisition, were outsourced to Artificial Intelligence. They considered such a future feasible given ongoing advancements in nanotechnology, which aims to create devices, systems, and materials at extremely small scales. At this scale, information could potentially be stored within nano-level structures, e.g., codes for multi-factor authentication. The miniature

[2] Solarpunk imagines a world were technology and nature work harmoniously with each other, instead of against each other

[3] Nanopunk describes a world where nanotechnology is the predominant technological force in society

scale in itself was seen as offering a form of security, as something so small may be difficult to access, breach, or tamper with. The likelihood of such a future is very high and to some extent it is already unfolding - just at a scale larger than nano.

> "*Maybe the table is actually really, really small, but some something about the structure of the really, really small table provide the security.*" [P4]

SOME LOUD ACCESSIBILITY EXTRACTS A DANGEROUS DESIGN

This sentence sparked a conversation about voice activation (hereafter VA) as an accessibility feature. However, participants noted that VA also introduces a variety of security risks, for instance, deepfaking a user's voice to gain access or to create phishing attempts. As a potential solution, participants suggested watermarking voices or applying cryptographic or hashing methods to VA systems. Others imagined a future in which VA for accessibility no longer exists, either because auditory senses have become overburdened and have lost some of their capabilities, or simply because VA is ultimately deemed too insecure to function as a reliable authentication method.

> *"Host: Could you secure your voice by any chance?*
> *P5: Some kind of watermark, perhaps."*

EXTRACT BLADE VIOLATES SECURE SHOT

This particular sentence sparked conversations about the protection of biometric data and broader information security concerns. Participants interpreted the "secure shot" as a kind of vaccine or capsule designed to protect a person's biometric data, while the "extract blade" was imagined as a countermeasure capable of neutralizing or destroying the effects of the vaccine. The secure shot was also discussed as a form of distress signalling, similar to firing a flare at sea in an emergency, but here applied to an individual signalling personal danger. The extract blade, as a countermeasure to the secure shot, was described as something that would ultimately compromise this security. Participants suggested that the motivation for deploying such a tool would likely be to gain access to sensitive information or to harm a user, driven primarily by financial goals.

> "*Maybe the secure shot is like the vaccine you can get and then it protects your biometric data. Then with the extract blades, you can you can extract the information anyway.*" [P5]

5 Vision and Future Work

The exquisite corpse exercise produces sentences in a serendipitous and unpredictable manner. The method offers considerable potential as a catalyst for far-fetched ideas, for instance, imagining trees as data centers or considering scale itself as a security mechanism. This may be attributed to the associative nature of language, as certain words can trigger multiple interpretations. For example, "secure shot" was understood both as a means of signaling danger (distress signaling) and as a means of preventing it (a vaccine shot). However, the value of these outputs largely depend on the deliberate cognitive work that follows, as participants are to make sense of the unexpected and translate it into coherent speculative insights. In workshop 1, the participants from the HCI domain readily interpreted the sentences and imagined far-fetched future scenarios without difficulties. Their discussions were rich and expansive, and this resulted in fewer sentences being produced. However, participants often drifted away from the workshop's central theme. In contrast, participants in workshop 2 – with technical security backgrounds – remained closely aligned with the central theme, but struggled to challenge current realities. Consequently, their discussions were more sparse, leading to the creation of more sentences. Thus, while this method allows participants to break free from conventional thinking and its limitations, its fullest potential is realized when individuals from different disciplines come together to turn randomness into relevance. This also supports the findings of [15].

For research the exquisite corpse method can serve as a powerful ideation tool to facilitate cutting-edge connections that would otherwise have been missed in conventional thinking. It enables researchers or the participants to explore possibilities and capabilities of a given topic by opening up a broader direction of possible and plausible domains. Beyond research context, the method also has practical value. It can be adopted by Research and Design (R&D) teams or biotech engineers, who are often tasked with exploring innovative ideas. Here, the technique can function both as a driver of creativity and as a means of speculating about the unintended consequences of the innovations. Similarly, the method can be used by municipalities and governmental ministries when planning large-scale technological rollouts, before committing to costly deployments.

To further develop this method, future work could conduct workshops with larger sample sizes and participants from diverse backgrounds and varying levels of familiarity with the case study – or apply it to other domains. In addition, more formalized analytical approaches could be employed to comparatively assess the potentials and limitations of the exquisite corpse as a speculative design method. Selected ideas could also be materialized into prototypes or provotypes to explore how speculative concepts translate into practice [3].

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